

Query

In the expression $((x+y+z)^{2026} + (x-y-z)^{2026})$, the brackets were expanded and like terms combined. How many terms of the form $(x^a y^b z^c)$ with a non-zero coefficient resulted?

Reasoning Summary

1. The expression $((x+y+z)^{2026} + (x-y-z)^{2026})$ will have non-zero coefficients for terms where the exponents of (y) and (z) sum to an even number.
2. The coefficient of each term $(x^a y^b z^c)$ in the sum is $((\frac{2026!}{a! b! c!} [1 + (-1)^{b+c}]))$. This coefficient is non-zero only if $(1 + (-1)^{b+c}) \neq 0$, which means $(b+c)$ must be even.
3. The problem reduces to counting the number of terms where $(a+b+c=2026)$ and $(b+c)$ is even. Since $(a=2026-(b+c))$ and 2026 is even, (a) is even if and only if $(b+c)$ is even.
4. Therefore, we need to count the number of non-negative integer solutions to $(a+b+c=2026)$ where (a) is even. This is equivalent to counting the number of solutions where $(b+c)$ is even.
5. For each even (a) , the number of solutions to $(b+c=2026-a)$ where $(b+c)$ is even is $((\frac{2026-a}{2} + 1))$. Summing this over all even (a) from 0 to 2026 gives $((1014)^2)$.
6. Hence, the number of terms with non-zero coefficients is $((1014)^2 = 1,028,196)$.

Final Answer

To determine the number of terms of the form $(x^a y^b z^c)$ with non-zero coefficients in the expression $((x+y+z)^{2026} + (x-y-z)^{2026})$, we analyze the expansion and combination of terms.

1. **Expansion Analysis**:

- The general term in $((x+y+z)^{2026})$ is $((\frac{2026!}{a! b! c!} x^a y^b z^c))$.
- The general term in $((x-y-z)^{2026})$ is $((\frac{2026!}{a! b! c!} x^a (-1)^{b+c} y^b z^c))$.

[... omitted for space ...]

5. **Summing Solutions**:

- The total number of solutions is the sum of $((\frac{2026-2k+1}{2}))$ for (k) from 0 to 1013.
- This sum is equivalent to summing $((2t+1))$ for (t) from 0 to 1013, which results in $((1014)^2)$.

6. **Conclusion**:

- The number of terms with non-zero coefficients is $((1014)^2 = 1028196)$.

R1 Ground Truth Reasoning Trace

Okay, so I have this problem here: In the expression $((x+y+z)^{2026} + (x-y-z)^{2026})$, after expanding and combining like terms, how many terms of the form $(x^a y^b z^c)$ have a non-zero coefficient? Hmm, interesting. Let me try to work through this step by step.

First, I remember that when you expand a trinomial like $((x+y+z)^n)$, the number of terms before combining is a combination with repetition problem. But in this case, we are combining like terms, so each term is determined by the exponents of x , y , and z . The general term in the expansion of $((x+y+z)^{2026})$ would be something like $((\frac{2026!}{a! b! c!} x^a y^b z^c))$ where $(a+b+c=2026)$. Similarly, for $((x-y-z)^{2026})$, each term would be $((\frac{2026!}{a! b! c!} x^a (-y)^b (-z)^c)) = ((\frac{2026!}{a! b! c!} x^a (-1)^{b+c} y^b z^c))$.

So when we add these two expressions together, the coefficients of each term $(x^a y^b z^c)$ would be $((\frac{2026!}{a! b! c!} [1 + (-1)^{b+c}]))$. Therefore, the coefficient will be non-zero only if $(1 + (-1)^{b+c}) \neq 0$. That happens when $((-1)^{b+c} = 1)$, which means that $(b+c)$ must be even. Therefore, in the sum, the terms where $(b+c)$ is even will have their coefficients doubled, and those where $(b+c)$ is odd will cancel out and have zero coefficient.

So the problem reduces to counting the number of monomials $(x^a y^b z^c)$ in the expansion where $(a+b+c=2026)$ and $(b+c)$ is even. Since $(a=2026-b-c)$, if $(b+c)$ is even, then (a) is even as well because 2026 is even. Wait, 2026 is even, right? Yes, 2026 divided by 2 is 1013, so even. Therefore, if $(b+c)$ is even, then $(a=2026-(b+c))$ is even. So all exponents (a) will be even, and $(b+c)$ will be even as well.

But actually, the key condition is just $(b+c)$ even, which in turn makes (a) even, since $(a=2026-(b+c))$. So the problem is equivalent to counting the number of non-negative integer solutions $((a,b,c))$ to $(a+b+c=2026)$ with $(b+c)$ even.

[... reasoning steps omitted for space ...]

Therefore, for each even m , the total number of solutions is $m+1$, which includes both cases where b and c are both even and both odd.

Therefore, when we compute the total number of terms, we are counting all such cases where $b+c$ is even, regardless of whether b and c are individually even or odd. Therefore, our original conclusion that the number of terms is $(1014)^2$ is correct.

But let's revisit the initial intuition where one might think that only terms with b and c even would survive. But in reality, as shown in the example with $n=2$, the term yz (with exponents 1 and 1) survives because $b+c=2$, which is even, even though b and c are both odd. Therefore, both cases contribute.

Therefore, the error in that initial intuition was assuming that both exponents must be even, but actually, it's the sum that must be even. Hence, both exponents even or both odd. Therefore, the correct count is indeed the number of solutions with a even, which is $((n/2 + 1)^2)$.

Therefore, after thorough analysis, the number of terms with non-zero coefficients in the given expression is $((1014)^2 = 1,028,196)$.

Final Answer The number of terms is $\boxed{1028196}$.

Reasoning Trace Synthesized by Our Inversion Model

Okay, so I have this problem here: I need to find the number of terms with non-zero coefficients in the expansion of $(x+y+z)^{2026} + (x-y-z)^{2026}$. Hmm, let's see. Alright, first, I remember that expanding a multinomial like $(x+y+z)^n$ can result in a lot of terms, each corresponding to different exponents of x , y , z . But since we're adding another similar expansion with some signs changed, maybe some terms will cancel out, and others will add up. So the key here is to figure out which terms survive after combining like terms.

Let me recall the general approach for expanding such expressions. When you expand $(x+y+z)^n$, each term is of the form $C(n, a, b, c) \cdot x^a y^b z^c$ where $a+b+c=n$, and $C(n, a, b, c)$ is the multinomial coefficient $n! / (a! b! c!)$. Similarly, for $(x-y-z)^n$, the coefficients would be the same but with alternating signs based on the number of negative terms in each term. Wait, actually, in the expansion of $(x-y-z)^n$, each term is $C(n, a, b, c) \cdot x^a (-y)^b (-z)^c = C(n, a, b, c) \cdot x^a (-1)^{b+c} y^b z^c$. So when you add the two expansions together, the coefficients for each term $x^a y^b z^c$ become $C(n, a, b, c) \cdot [1 + (-1)^{b+c}]$.

Therefore, the coefficient will be non-zero only if $1 + (-1)^{b+c} \neq 0$, which means $(-1)^{b+c} \neq -1$, so $(-1)^{b+c} = 1$. That implies that $b+c$ must be even. Because if $b+c$ is even, then $(-1)^{b+c} = 1$, so the coefficients add up. If $b+c$ is odd, they cancel out. So, in the final expression, only the terms where $b+c$ is even will have non-zero coefficients. Therefore, the number of terms we are looking for is the number of triples (a, b, c) such that $a+b+c=2026$ and $b+c$ is even.

Wait, but $a=2026-(b+c)$, so substituting that in, we have $2026-(b+c)+b+c=2026$, which just checks out. So the condition is that $b+c$ is even. Therefore, the problem reduces to counting the number of non-negative integer solutions (a, b, c) to $a+b+c=2026$ where $b+c$ is even. Since a is determined by b and c ($a=2026-(b+c)$), the condition on a would automatically be satisfied because $a=2026-(b+c)$, and 2026 is even. Wait, 2026 is an even number.

[... reasoning steps omitted for space ...]

So four terms, which matches the formula $((3+1)/2)^2 = 4$. So my initial mistake was in the manual calculation where I thought there were three terms, but actually, xyz terms also contribute. So for $n=3$, the number of terms is 4. Therefore, the formula holds.

So the general formula is: if n is even: number of terms = $((n/2 + 1)^2)$ if n is odd: number of terms = $((n+1)/2)^2$

Wait, but in our original problem, $n=2026$, which is even. So the number of terms is $((2026/2 + 1)^2 = (1013+1)^2 = 1014^2 = 1,028,196)$.

Therefore, the answer should be 1,028,196.

But let me verify this once more with another example. Take $n=4$, even. Number of terms = $((4/2+1)^2 = 3^2 = 9)$. As before, correct. For $n=6$, $((6/2+1)^2 = 4^2 = 16)$, which matches.

Therefore, applying this formula to our original problem gives the answer. Hence, the number of terms with non-zero coefficients is $(1014)^2 = 1,028,196$.

Final Answer $\boxed{1028196}$.